

1. Administrative & Study Identification

Full Study Title: A Phase Ib/II, Multicenter Study of the Combination of LEE011 and BYL719 With Letrozole in Adult Patients With Advanced ER+ Breast Cancer **Short Title/Acronym:** Study of LEE011, BYL719 and Letrozole in Advanced ER+ Breast Cancer
Protocol Number: CLEE011X2107 **NCT Number:** NCT01872260 **Posting ID:** 782986010198506109 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** Novartis **Investigational Product:** LEE011 (ribociclib), BYL719 (alpelisib), and Letrozole (Trade Name: Femara, ATC Code: L02BG04) **Phase of Development:** Phase I/Phase II (PHASE1, PHASE2) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To estimate the Maximum Tolerated Dose (MTD) and/or Recommended Phase II Dose (RP2D) for three regimens: LEE011 with letrozole, BYL719 with letrozole, and the triple combination of LEE011 + BYL719 with letrozole.
Secondary Objectives: To further characterize the safety, tolerability, pharmacokinetics (PK), and preliminary clinical anti-tumor activity (including Overall Response Rate) of the combinations during the dose expansion phase.

2.2 Background & Rationale Activation of the PI3K/AKT/mTOR and cyclin D-CDK4/6-INK4-Rb pathways has been heavily implicated in endocrine therapy resistance in patients with advanced hormone-receptor-positive (HR+) breast cancer. Preclinical models demonstrated that combining LEE011 (a CDK4/6 inhibitor), BYL719 (a PI3K-alpha inhibitor), and letrozole yields enhanced antitumor activity compared to either agent alone. This trial aims to inform the future clinical development of these investigational targeted agents in ER+ breast cancer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm/Parallel assignment (Multiple escalating dose arms without placebo control) **Blinding:** Open-label **Randomization:** Non-randomized/Parallel Assignment

3.2 Planned Enrollment & Duration Target N\$: 255 adult women
Number of Sites: Multicenter **Estimated Study Start:** 22-Oct-2013
Estimated Primary Completion: 26-Feb-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult women (18-100 years of age) with postmenopausal, Estrogen-receptor (ER) positive and/or Progesterone-receptor positive breast cancer. 2. Phase Ib dose escalation only: Any number of prior lines of endocrine therapy is allowed, except for cytotoxic therapy, which is limited to one prior line administered in the advanced (metastatic or locally advanced) setting. 3. Phase Ib dose expansions (Arms 1, 2, 3): No prior systemic treatment in the advanced setting, with the exception of treatment with letrozole for a maximum of one month prior to starting study treatment. 4. Patients who received (neo)adjuvant therapy for breast cancer are eligible. Prior therapy with letrozole or anastrozole in the (neo)adjuvant setting is permitted if the disease-free interval is greater than 12 months from the completion of treatment.

4.2 Exclusion Criteria 1. HER2-overexpression in the patient's tumor tissue. 2. Patients with active CNS or other brain metastases. 3. Clinically manifest diabetes mellitus (treated and/or clinical signs or with fasting glucose ≥ 126 mg/dL / 7.0 mmol/L or hemoglobin A1c $\geq 6.5\%$), history of gestational diabetes, or documented steroid-induced diabetes. 4. Major surgery within 2 weeks, acute or chronic pancreatitis, bilateral diffuse lymphangitic carcinomatosis, or another malignancy within 3 years. 5. Receiving hormone replacement therapy that cannot be discontinued. 6. Impaired cardiac function.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm 1:** LEE011 (dose escalating, 28-day cycles: 21 days on, 7 days off) + letrozole (2.5 mg/day). * **Arm 2:** BYL719 (dose escalating, daily continuous) + letrozole (2.5 mg/day). * **Arm 3:** LEE011 (dose escalating, 21 days on, 7 days off) + BYL719 (dose escalating, daily) + letrozole (2.5 mg/day). * **Arm 4:** LEE011 (dose escalating, daily continuous) + BYL719 (dose escalating, daily) + letrozole (2.5 mg/day).

Route of Administration: Oral **Duration of Treatment:** 28-day cycles administered until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant supportive care medications not metabolically conflicting with the investigational drugs. **Prohibited:** Hormone replacement therapy that cannot be discontinued; CYP3A4 modifier drug treatments ≤ 2 weeks before starting treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Tumor Assessments, Labs
Baseline	Day 0	Treatment assignment, First Dose, PK Evaluation
Interim	Every 28 days	Cycle 1-X Onsite Visits: Safety Labs, DLT Assessment, PK testing, Efficacy Assessment (RECIST)
End of Study	Follow-up	Final Vital Signs, Exit Interview, Disease Progression confirmation, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Incidence of Dose limiting toxicities (DLTs)** - Phase Ib only.
- 2. Safety and tolerability:** Adverse Events (AEs), serious AEs (SAEs), changes in hematology and chemistry values, vital signs, electrocardiograms (ECGs), dose interruptions, reductions and dose intensity.
- 3. PK profiles of LEE011 and letrozole:** To characterize PK profiles of LEE011 and Letrozole.

7.2 Secondary & Exploratory Endpoints

- 1. Safety and tolerability:** Of LEE011 in combination with letrozole, BYL719 in combination with letrozole, and the triple combination of LEE011 + BYL719 with letrozole.
- 2. Plasma concentration-time profiles:** Of LEE011, BYL719 and letrozole.
- 3. Overall Response Rate (ORR):** Preliminary clinical anti-tumor activity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical exams, cardiac function evaluations (ECGs), and clinical lab visits continuously through the treatment cycles.

Serious Adverse Events (SAEs): Routine SAE reporting timelines consistent with Phase I oncology standards.

Data Monitoring Committee (DMC): Utilizes a Bayesian Logistic Regression Model (BLRM) with the escalation with overdose control principle and real-time PK assessments to guide safe dose escalations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety set (all subjects receiving ≥ 1 dose of study drug); Dose-determining set (evaluable for DLTs in Cycle 1).

Sample Size Justification: Target enrollment of 255 patients, dynamically guided by the BLRM for dose escalation cohorts to confidently estimate the MTD.

Handling of Missing Data: Subjects discontinuing before completing 21 days in Cycle 1 for reasons other than a DLT are considered non-evaluable for MTD determination and are typically replaced.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous monitoring by the sponsor; institutional ethics committee and regulatory authority oversight at participating institutions. **Audit Trail:** Standard clinical trial electronic Case Report Form (eCRF) locks, source data verification, and data clarification mechanisms.

11. Study Identifiers & Governance

Posting ID: 782986010198506109 **Primary ID:** CLEE011X2107 **Registry Number:** NCT01872260 **Sponsor/Lead Organization:** Novartis **Study Title:** Study of LEE011, BYL719 and Letrozole in Advanced ER+ Breast Cancer **Official Regulatory Title:** A Phase Ib/II, Multicenter Study of the Combination of LEE011 and BYL719 With Letrozole in Adult Patients With Advanced ER+ Breast Cancer **Trial Status:** ACTIVE_NOT_RECRUITING

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Breast Cancer **Preferred UMLS Name:** Estrogen Receptor Positive Breast Neoplasms **Diagnosis Threshold:** Postmenopausal, Estrogen-receptor (ER) positive and/or Progesterone-receptor (PR) positive, HER2-negative **Medical Methodology:** CDK4/6 and/or PI3K- α Targeted Inhibitors combined with Endocrine Therapy **Optimization Target:** Maximum Tolerated Dose (MTD) / Recommended Phase II Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Dose escalation followed by dose expansion clinical pathway **Education Protocol (HCP):** Dose-limiting toxicity (DLT) criteria and BLRM-guided escalation rules **Education Protocol (Patient):** Strict daily oral dosing schedules and safety reporting **Quality Audit Mechanism:** Real-time Pharmacokinetic (PK) tracking and continuous safety evaluation

14. Observation & Follow-Up Schedule

Total Duration: Monitored continuously until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Every 28-day cycle interval **Remote Monitoring Frequency:** Routine AE follow-up post-discontinuation **Checklist Review Interval:** Regular tumor assessments per RECIST guidelines

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					